M460 Applied Data Mining

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Course Webpage

Applied Data Mining

Office Hours:

Office: F2409 Unistructure

WF: 12pm-2pm or by appointment

Zoom's Link

Course Description

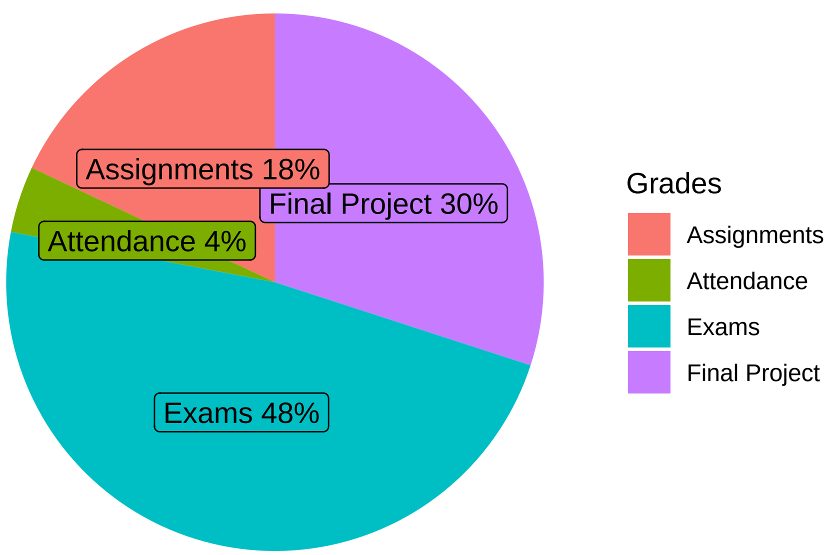
This course provides extensive hands-on experience with SAS Enterprise Miner software and covers the basic skills required to assemble analyses using the rich tool set of both packages. It also covers concepts fundamental to understanding and successfully applying data mining methods.

Course Objectives

After completing this course, you should be able to

- prepare data for analysis, including partitioning data and imputing missing values
- explain the algorithms of common predictive models such as decision trees, regression models, and neural networks.
- train, assess, and compare multiple predictive models such as decision trees, regression models, and neural networks.

Grades



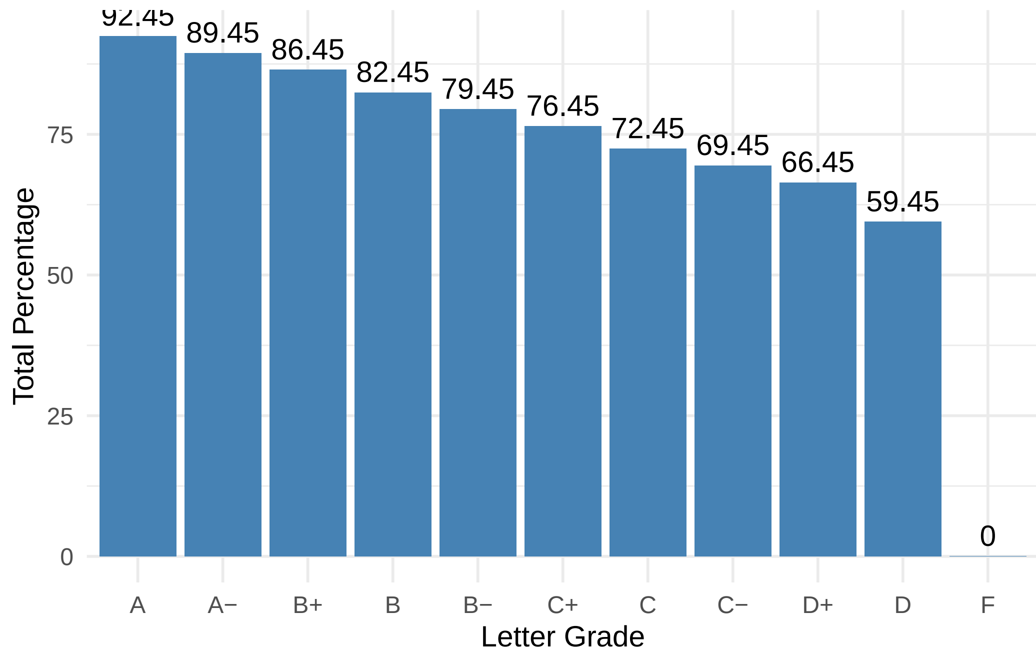
- *Final Project*: The final project offers you an opportunity to apply data mining techniques to an application. Projects can be done in teams of up to three students. If you have a project of such large scope and ambition that it cannot be done by a team of only three persons, you can propose doing a project in a team of four. The logistics and other details of the project will be updated on the course webpage and Canvas.

- *Exams*: The exams are written exams. Calculators and a formula sheet are allowed. There will be practice problems for each exam to help you prepare for the exam.
- *Assignments*: The assignments include, but not limited to, solving statistical problems, writing reports about a statistical technique, and presenting statistical analysis on a dataset.
- *Attendance*: Attendance will be checked regularly in class. Missing fewer than four class meetings will guarantee you the full credits for attendance. The grade for attendance (4%) is enough to move up a letter grade. You are responsible for obtaining course material missed during an absence.

Grades Scale

The numerical grades are converted to letter grades as follows.

A	92.45 - 100%	C	72.45 - 76.44%
A-	89.45 - 92.44%	C-	69.45 - 72.44%
B+	86.45 - 89.44%	D+	66.45 - 69.44%
B	82.45 - 86.44%	D	59.45 - 66.44%
B-	79.45 - 82.44%	F	Below 59.44%
C+	76.45 - 79.44%		



Late Work

Late assignments are penalized 10% for each day late. You can resubmit your work to improve your score, but the late penalty will apply.

Tentative Topics

- Overfitting and Model Tuning
- Classification Tree
- Random Forest
- Adaboost and Boosting Algorithm
- Gradient Boosting
- K-Nearest Neighbors
- Recommendation System
- ElasticNet
- Variable selection
- Neural Network and Deep Learning

Academic Misconduct

Cheating will result in an "F" as your final grade and may result in your expulsion from the University. All cheating will be reported to the Chair of the Mathematics Department and Academic Advising.

Regarding Diversity

In this course, and all your courses at Bryant, and throughout the Bryant learning community, we value and respect diversity. This includes differences in race, ethnicity, nationality, gender, gender identity, sexuality, socioeconomic status, ability, and religion.

Honor Contract

1. Honors Deliverables & Scope

The Honors student will investigate state-of-the-art methodology in machine learning on imbalanced datasets, surveying theoretical formulations, algorithmic paradigms (e.g., re-sampling strategies, cost-sensitive learning, ensemble mechanisms), and evaluation metrics suited for severe class skew (e.g., PR-AUC, G-Mean, F-beta).

- **Milestone 1: Research Proposal & Annotated Bibliography (5%)**
  - Due: End of Week 6.
  - A 2-page research outline defining the taxonomy of imbalanced learning techniques to be surveyed, accompanied by an annotated bibliography of at least 8–10 peer-reviewed journal or conference papers (e.g., IEEE, ACM, NeurIPS, ICML, KDD).
- **Milestone 2: Comprehensive Literature Review Manuscript (15%)**
  - Due: End of Week 12.
  - An 8–12 page survey paper critically synthesizing the literature. The review must not merely summarize individual papers sequentially, but instead:
    - Categorize techniques across data-level, algorithm-level, and hybrid/ensemble paradigms.
    - Compare inductive biases, computational overhead, and empirical failure modes (e.g., borderline noise sensitivity, small disjuncts).
    - Highlight recent advances and open challenges.
- **Milestone 3: Seminar Presentation & Oral Defense (10%)**

- Due: Weeks 13–14 (scheduled in consultation with the instructor).
- A 20-minute slide presentation delivered to the class or research seminar, followed by a 10-minute Q&A/oral defense. The student will walk through the core taxonomy, analyze key experimental benchmarks from the surveyed literature, and respond to methodological questions.

2. Honors Grading Criteria

The Honors deliverables will be evaluated against academic research standards using the following criteria:

- **Depth & Critical Analysis (20%):** Going beyond surface summaries to compare methods, highlight theoretical trade-offs, and identify methodological weaknesses across studies.
- **Taxonomy & Organization (20%):** A logical, clearly structured framework that effectively categorizes and contrasts different imbalanced learning algorithms.
- **Technical Rigor & Accuracy (20%):** Accurate mathematical definitions and explanations of sampling formulas, objective/loss functions, and evaluation metrics.
- **Scholarly Writing & Citation Quality (10%):** Clean academic prose, adherence to standard publication formatting, and thorough, correctly formatted citations.
- **Oral Communication & Defense (30%):** Clear slide design, effective delivery within time limits, and the ability to confidently defend methodological choices during Q&A.